FORMULA MAP · START WITH THE ENERGY STORY

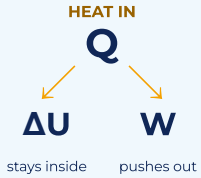
Heat in. *Energy kept.* Work out.

One law connects everything. Find what stays fixed, then fill in the zeros.

THE FIRST LAW

$$Q = \Delta U + W$$

Heat received = energy stored + work done by the gas.



Keep the signs

Heat enters: $Q > 0$
Expansion: $W > 0$
Compression: $W < 0$

THE GAS LAW

$$PV = nRT$$

ENERGY STORED

$$U = \frac{f}{2}nRT = \frac{f}{2}PV$$

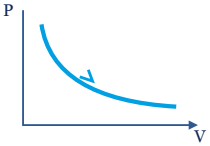
ANY IDEAL-GAS PROCESS

$$\Delta U = nC_v\Delta T$$

Same endpoints → same ΔU.

T Isothermal

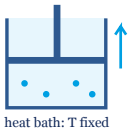
T stays fixed



$$\Delta U = 0 \quad Q = W = nRT\ln\frac{V_2}{V_1}$$

$PV = \text{constant}$ · $W = nRT\ln(P_1/P_2)$

Same temperature → same U. Heat becomes work. $C \to \infty$ when heat flows.



P Isobaric

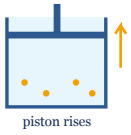
P stays fixed



$$W = P\Delta V = nR\Delta T \qquad Q = nC_p\Delta T$$

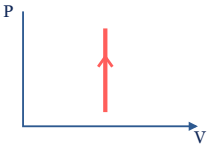
$V/T = \text{constant}$ · $C = C_p$

Free piston → heat warms AND pushes. $Q : \Delta U : W = C_p : C_v : R$



V Isochoric

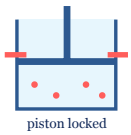
V stays fixed



$$W = 0 \qquad Q = \Delta U = nC_v\Delta T$$

$P/T = \text{constant}$ · $C = C_v$

Locked piston → no movement → no work. All the heat stays inside.



◆ Heat capacity shortcuts

$$C_v = \frac{f}{2}R \quad C_p = C_v + R \quad \gamma = \frac{C_p}{C_v} = 1 + \frac{2}{f}$$

$$C_v = \frac{R}{\gamma - 1} \qquad C_p = \frac{\gamma R}{\gamma - 1}$$

Gas	f	C_v/R	C_p/R	γ
One atom	3	3/2	5/2	5/3
Two atoms	5	5/2	7/2	7/5
Nonlinear rigid	6	3	4	4/3

$C = \delta Q/(n\,dT)$; for constant C, $Q = nC\Delta T$.

◆ Mixtures: each mole gets a vote

$$C_{v,\text{mix}} = \frac{n_1C_{v1} + n_2C_{v2}}{n_1 + n_2}$$

$$f_{\text{mix}} = \frac{n_1f_1 + n_2f_2}{n_1 + n_2} \qquad \gamma_{\text{mix}} = 1 + \frac{2}{f_{\text{mix}}}$$

$C_{p,\text{mix}} = C_{v,\text{mix}} + R$. **Never average γ.**
Equal moles, f = 3 and 5 → $f_{\text{mix}} = 4$, $\gamma_{\text{mix}} = 3/2$.

At constant P: $\Delta U/Q = 1/\gamma$; $W/Q = (\gamma - 1)/\gamma$.

Water, metal & phase change

$Q = mc\Delta T$ or mL; $W = P\Delta V$ (P fixed)
$V = m/\rho$; $\Delta U = Q - W$. Heat-input rate − work-output rate = dU/dt.

Read the symbols

n = moles; f = degrees of freedom; C is per mole.
1 = initial, 2 = final; Δ = final − initial.
U is a state function; Q and W depend on the path.

ADIABATIC CHANGES · FREE EXPANSION · CYCLES

Spend stored energy. *Close the loop.*

No heat? The gas uses its own energy. Back to the start? The energy change is zero.

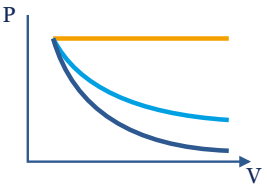
Q Adiabatic: no heat in or out

Q = 0

$$W = -\Delta U = nC_v(T_1 - T_2)$$

$$W = \frac{nR(T_1 - T_2)}{\gamma - 1} = \frac{P_1V_1 - P_2V_2}{\gamma - 1}$$

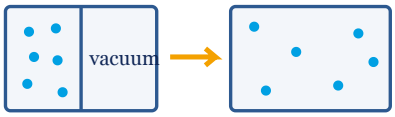
$PV^\gamma = \text{const}$ → $TV^{\gamma-1} = \text{const}$ → $P \propto T^{\gamma/(\gamma-1)}$



— Isobaric — Isothermal
— Adiabatic

Expansion does work → T falls. Compression → T rises.
$C = 0$. The three relations above require a reversible adiabatic path.

Same initial state + same final V: expansion $W_P > W_T > W_{ad}$. Compression: reverse the order for $|W|$.
At the same P,V: slopes are $-P/V$ (isothermal) and $-\gamma P/V$ (adiabatic).



Remove the partition. Nothing gets pushed.

Free expansion: no one to push

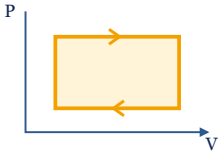
$$Q = W = \Delta U = 0 \quad \Delta T = 0$$

Insulated box + vacuum + ideal gas. $P_1V_1 = P_2V_2$ at the endpoints.
No equilibrium P–V path. Do not use $PV^\gamma = \text{const}$.

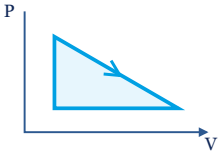
THERMODYNAMIC CYCLES · BACK TO THE SAME STATE

$$\Delta U_{\text{cycle}} = 0 \rightarrow Q_{\text{net}} = W_{\text{net}} = \oint P\,dV$$

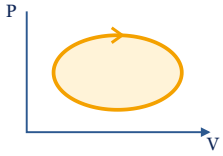
Net work = signed area inside the loop. Clockwise is positive when P is vertical and V is horizontal.



Rectangle
$|W| = \Delta P\,\Delta V$



Triangle
$|W| = \frac{1}{2}\Delta P\,\Delta V$



Circle / ellipse
$|W| = \frac{\pi}{4}\Delta P\,\Delta V$

Read direction first

Clockwise: $+W$
Anticlockwise: $-W$

ΔP and ΔV are **full spans**.
Swap axes → reverse signs.
Several loops → add signed areas.

◆ Long cycle? Make a table.

Name each step → fill the zeros → $Q = \Delta U + W$.

Step	Q / J	ΔU / J	W / J
AB · T fixed	+700	0	+700
BC · Q = 0	0	−400	+400
CA · return	−100	+400	−500
Cycle	**+600**	**0**	**+600**

Check: $\sum \Delta U = 0$; $\sum Q = \sum W$.
CA: 500 J of work is done **on** the gas.

◆ Heat in is not net heat

Q_{in} = sum of positive Q's.
Q_{out} = magnitude of negative Q's added together.

$$Q_{\text{net}} = Q_{\text{in}} - Q_{\text{out}} = W_{\text{net}}$$

Engine preview $\eta = \frac{W_{\text{net}}}{Q_{\text{in}}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}}$

Power = work per cycle × cycles per second.
N cycles/min → $\text{Power} = NW_{\text{net}}/60$.

Other paths
$W = \int P\,dV$; straight line: $W = \frac{1}{2}(P_1 + P_2)(V_2 - V_1)$.
Curved P(V)? Integrate. P–T area is not work.
$W_{\text{on}} = -W_{\text{by}}$.

Units save marks
Pa·m³ = J; kPa·L = J; kPa·cm³ = 10^{-3} J.
L·atm ≈ 101.3 J; R = 8.31 J mol^{-1} K^{-1}.
Use kelvin: T(K) = t(°C) + 273.15.

Ideal gas, fixed n and constant heat capacities. Gas-pressure work paths assume reversible processes. Zeroth law: $T_A = T_C$, $T_B = T_C \Rightarrow T_A = T_B$.